

MediCare — Phase 6: User Interface Development

Introduction: This phase focuses on building an intuitive Lightning experience for different hospital roles (Doctor, Receptionist, Admin). I used Lightning App Builder, Record Pages, Tabs, Home Page Layouts, Utility Bar, a custom LWC and Apex integration to deliver a seamless UI.

1. Lightning App Builder

To provide different users with customized pages, I used Lightning App Builder to design **Appointment – Doctor View** and **Appointment – Receptionist View** record pages. Each page shows only the components relevant to that role.

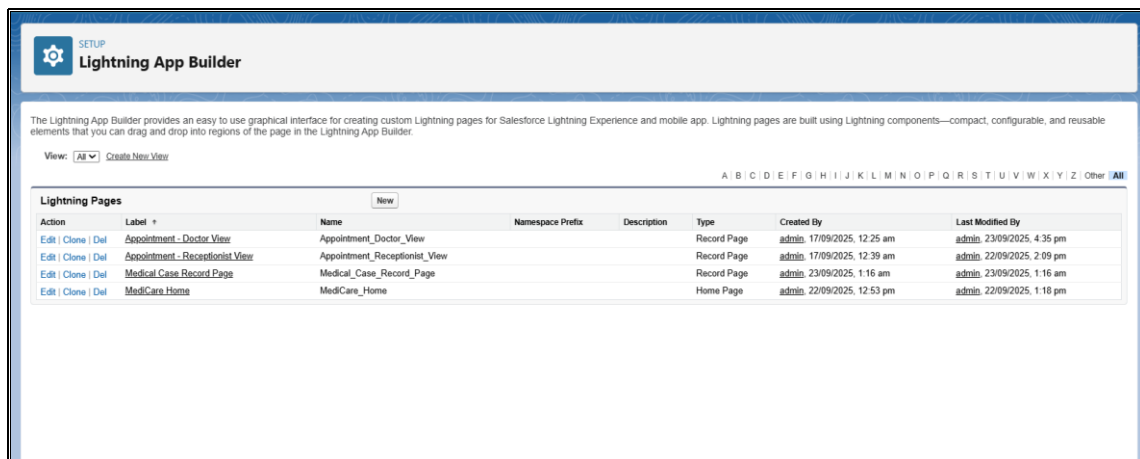


Fig-1 Lightning App Builder - Lightning Pages

2. Record Pages (Doctor & Receptionist Views)

Doctors need quick access to upcoming appointments and related Medical Cases. Receptionists need fast appointment entry. I created two record pages with appropriate components:

- **Appointment – Doctor View:** shows patient summary, related Medical Cases, and the upcoming appointments LWC.
- **Appointment – Receptionist View:** optimized for quick data entry and follow-up creation.
- Activated per profile/record type so each user sees the correct page.

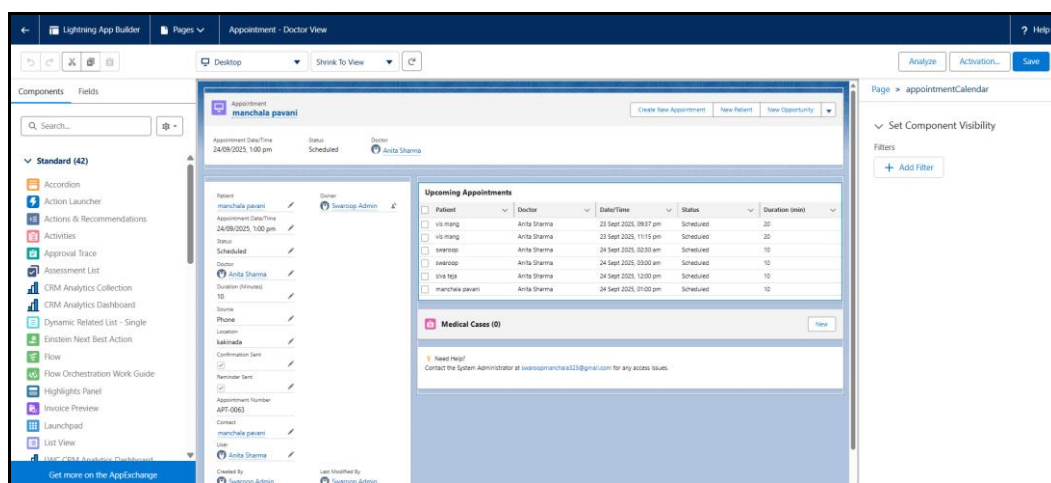


Fig-2.1 Appointment – Doctor View (Record Page)

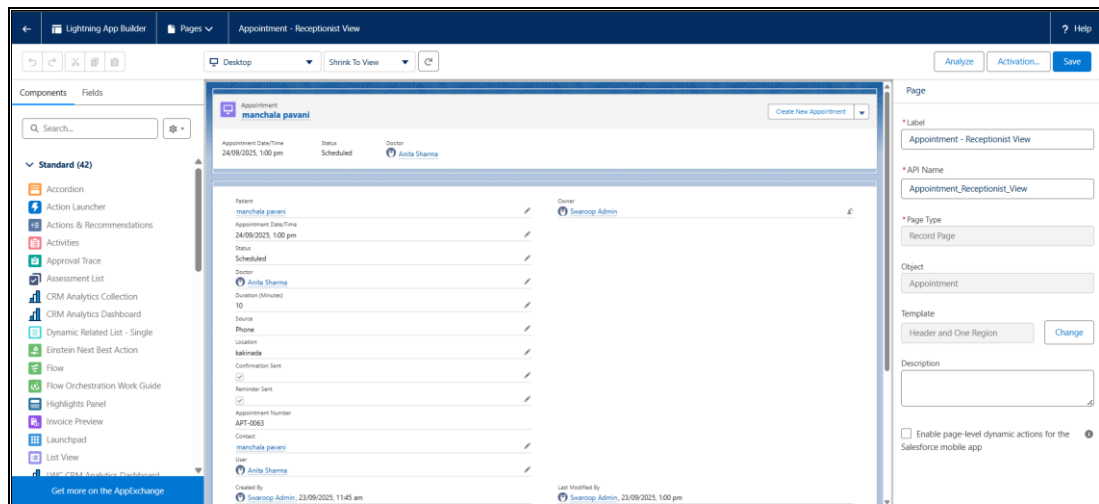


Fig-2.2 Appointment – Receptionist View (Record Page)

3. Tabs

I exposed all major objects (Patients, Appointments, Medical Cases) as tabs in the MediCare Lightning app so staff can navigate quickly.

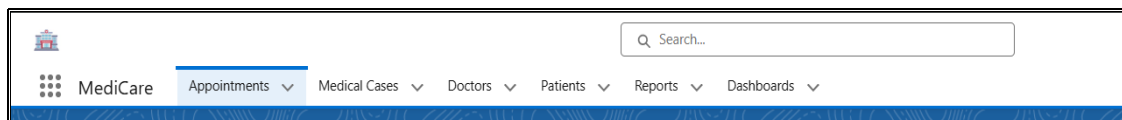


Fig-3 Tabs in App

4. Home Page Layouts

I built a custom **MediCare Home Page** that displays a dashboard component (Upcoming Appointments, Recent Appointments & Medical cases, Upcoming Appointments Report Chart) for a real-time view.

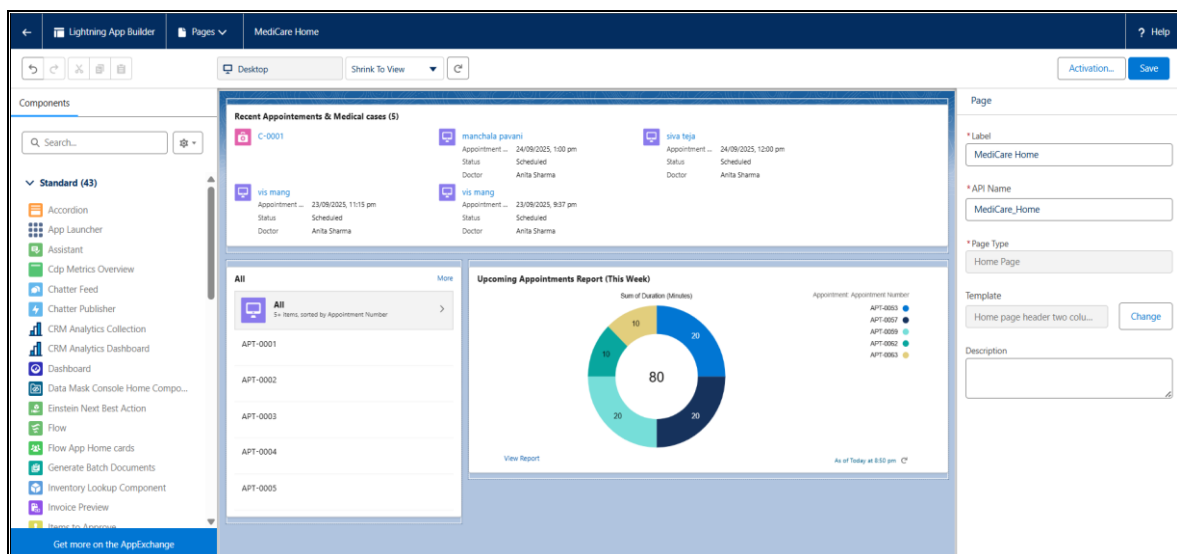


Fig-4 MediCare Home Page Layout

5. Utility Bar

I added Reports/Dashboards and Quick Actions to the Utility Bar to give staff one-click access from anywhere in the app.

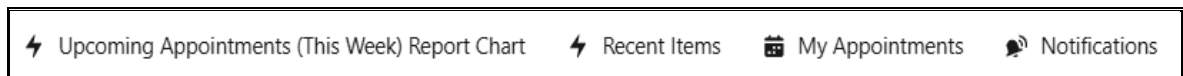


Fig-5.1 Utility Bar

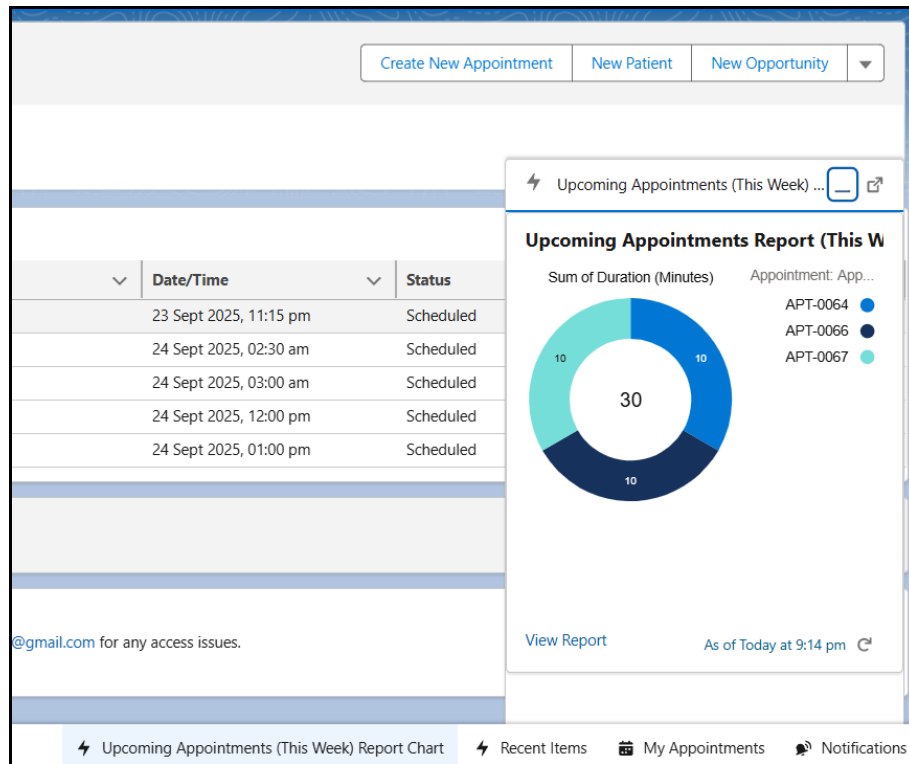


Fig-5.2 Testing Utility Bar

6. Lightning Web Component (LWC) – “Upcoming Appointments”

- A custom LWC appointmentCalendar built and deployed.
- It queries all upcoming appointments and displays Patient, Doctor, Date/Time, Status, Duration in a datatable.
- Dropped into the Appointment – Doctor View page so the doctor can instantly see upcoming appointments.

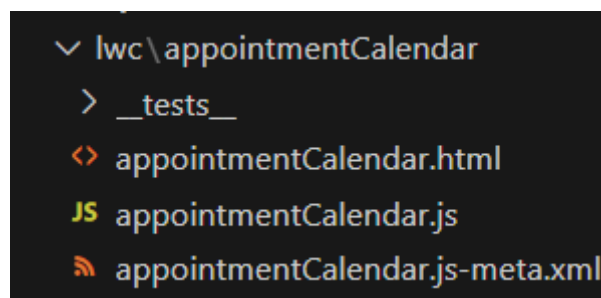


Fig-6.1 LWC Files Structure

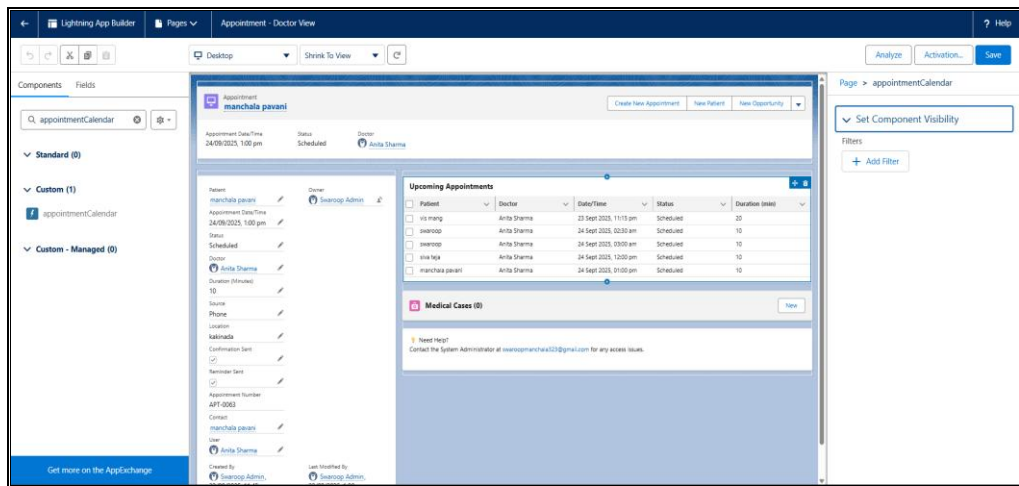


Fig-6.2 LWC_AppBuilder

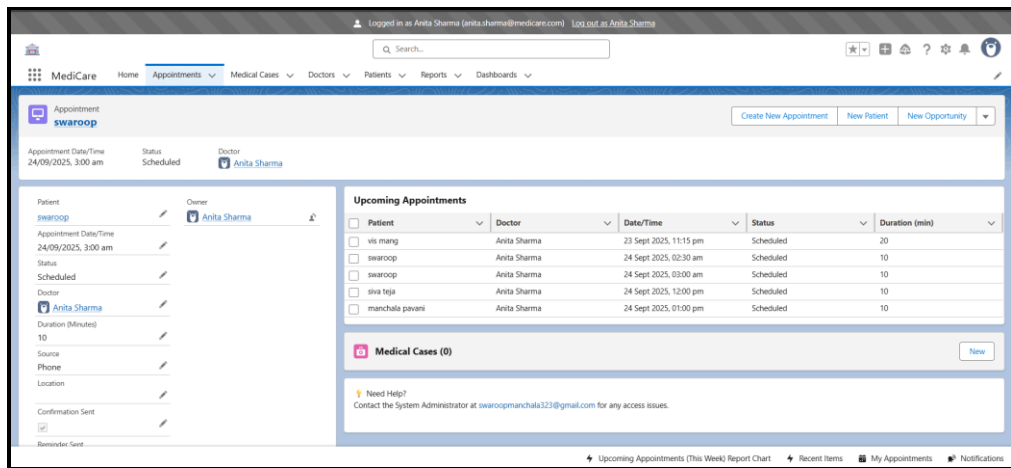


Fig-6.3 LWC_DoctorView

7. Apex with LWC

The LWC calls the Apex class AppointmentController using a wire adapter to fetch upcoming appointments server-side.

This ensures the doctor always sees real-time appointment data without page refresh.

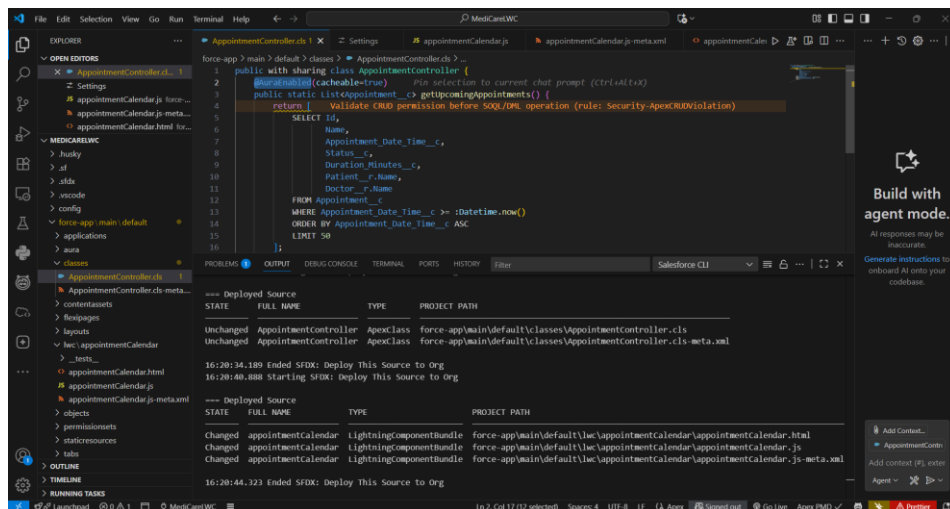
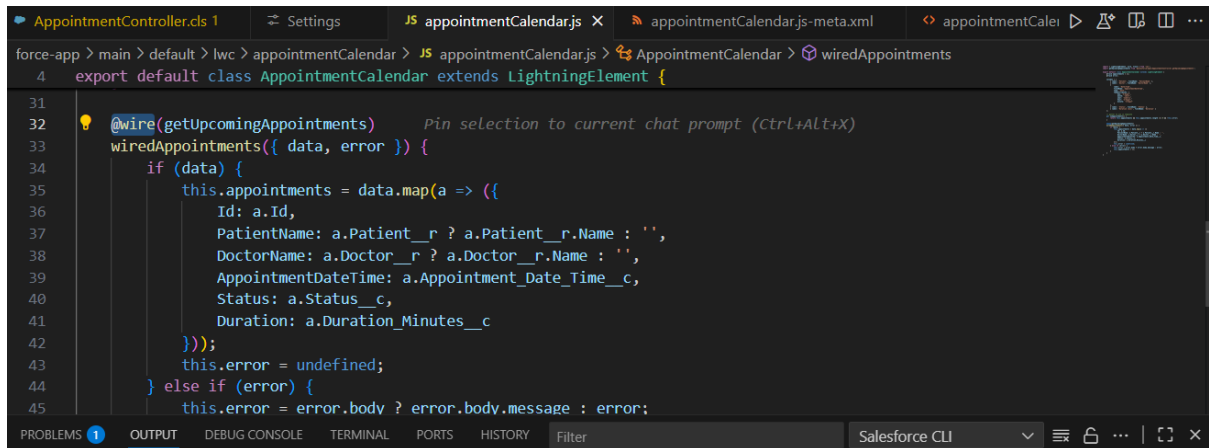


Fig-7 Apex With LWC

8. Wire Adapters

I used a **@wire** adapter in the LWC to call the Apex method automatically and refresh when the data changes.

This keeps the appointments list up-to-date without manual refresh.



```
AppointmentController.cls 1  Settings  JS appointmentCalendar.js  appointmentCalendar.js-meta.xml  appointmentCalendar.js  wiredAppointments
force-app > main > default > lwc > appointmentCalendar > JS appointmentCalendar.js > AppointmentCalendar > wiredAppointments
4  export default class AppointmentCalendar extends LightningElement {
31
32  @wire(getUpcomingAppointments)    Pin selection to current chat prompt (Ctrl+Alt+X)
33  wiredAppointments({ data, error }) {
34    if (data) {
35      this.appointments = data.map(a => ({
36        Id: a.Id,
37        PatientName: a.Patient__r ? a.Patient__r.Name : '',
38        DoctorName: a.Doctor__r ? a.Doctor__r.Name : '',
39        AppointmentDateTime: a.Appointment_Date_Time__c,
40        Status: a.Status__c,
41        Duration: a.Duration_Minutes__c
42      }));
43      this.error = undefined;
44    } else if (error) {
45      this.error = error.body ? error.body.message : error;
```

Fig-8 @wire adapter in LWC

9. Results / Observations

- **Doctors** now see a live list of **upcoming appointments** directly on their **Appointment** page.
- **Receptionists** can quickly create and manage appointments without leaving their view.
- The **MediCare Home** Page provides a real-time snapshot of the hospital's activities (appointments and cases).
- **Utility Bar** ensures quick access to **reports**, **dashboards** and **actions** from anywhere in the app.
- The **LWC** with **Apex integration** makes the **UI dynamic** and responsive, reducing page refreshes.